

Yinda Zhang

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EDUCATION

University of Pennsylvania

Advisor: Prof. Vincent Liu

Ph.D., Computer and Information Science

Sep. 2022 - Present

University of Chicago

Advisor: Prof. Junchen Jiang

Pre-Doctoral M.S., Computer Science

Sep. 2020 - Dec. 2021

Peking University

Advisor: Prof. Tong Yang

B.S., Computer Science and Technology

Sep. 2016 – Jul. 2020

PUBLICATION

Conference

1. **Yinda Zhang**, Liangcheng Yu, Gianni Antichi, Ran Ben Basat, Vincent Liu, Enabling Silent Telemetry Data Transmission with InvisiFlow, USENIX Symposium on Networked Systems Design and Implementation (NSDI 2025)
2. **Yinda Zhang**, Peiqing Chen, Zaoxing Liu, OctoSketch: Enabling Real-Time, Continuous Network Monitoring over Multiple Cores, USENIX Symposium on Networked Systems Design and Implementation (NSDI 2024)
3. Yikai Zhao*, Wenchen Han*, Zheng Zhong*, **Yinda Zhang**, Tong Yang, Bin Cui, Double-Anonymous Sketch: Achieving Fairness for Finding Global Top-K Frequent Items, ACM International Conference on Management of Data (SIGMOD 2023)
4. Yikai Zhao*, **Yinda Zhang***, Yuanpeng Li*, Yi Zhou, Chunhui Chen, Tong Yang, Bin Cui, MinMax Sampling: A Near-optimal Global Summary for Aggregation in the Wide Area, ACM International Conference on Management of Data (SIGMOD 2022)
5. Zhuochen Fan, **Yinda Zhang**, Tong Yang, Mingyi Yan, Gang Wen, Yuhan Wu, Hongze Li, Bin Cui, PeriodicSketch: Finding Periodic Items in Data Streams, IEEE International Conference on Data Engineering (ICDE 2022)
6. Tong Yang, Jizhou Li, Yikai Zhao, Kaicheng Yang, Hao Wang, Jie Jiang, **Yinda Zhang**, Nicholas Zhang, QCluster: Clustering Packets for Flow Scheduling, International World Wide Web Conference (WWW 2022)
7. **Yinda Zhang**, Zaoxing Liu, Ruixin Wang, Tong Yang, Jizhou Li, Ruijie Miao, Peng Liu, Ruwen Zhang, Junchen Jiang, CocoSketch: High-Performance Sketch-based Measurement over Arbitrary Partial Key Query, Annual conference of the ACM Special Interest Group on Data Communication (SIGCOMM 2021)
8. **Yinda Zhang**, Jinyang Li, Yutian Lei, Tong Yang, Zhetao Li, Gong Zhang, Bin Cui, On-Off Sketch: A Fast and Accurate Sketch on Persistence, International Conference on Very Large Data Bases (VLDB 2021)
9. Xiangyang Gou*, Long He*, **Yinda Zhang***, Ke Wang, Xilai Liu, Tong Yang, Yi Wang, Bin Cui, Sliding Sketches: A Framework using Time Zones for Data Stream Processing in Sliding Windows, ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD 2020)

Journal

1. Yikai Zhao, Wei Zhou, Wenchen Han, Zheng Zhong, **Yinda Zhang**, Xiuqi Zheng, Tong Yang, Bin Cui, Achieving Top-K-fairness for Finding Global Top-K Frequent Items, IEEE Transactions on Knowledge and Data Engineering (TKDE 2025)
2. Ruijie Miao*, **Yinda Zhang***, Zihao Zheng*, Ruixin Wang, Ruwen Zhang, Tong Yang, Zaoxing Liu, Junchen Jiang, High-Performance Sketch-based Measurement over Arbitrary Partial Key Query, IEEE/ACM Transactions on Networking (TON 2023)
3. Xiangyang Gou*, **Yinda Zhang***, Zhoujing Hu*, Long He, Ke Wang, Xilai Liu, Tong Yang, Yi Wang, Bin Cui, A Sketch Framework for Approximate Data Stream Processing in Sliding Windows, IEEE Transactions on Knowledge and Data Engineering (TKDE 2022)

SELECTED RESEARCH PROJECTS

Programmable Networks

Collecting Network Telemetry Without Affecting User Traffic Nov. 2022 – Sep. 2024

- Developed a communication substrate for silently collecting network telemetry from switches
- Implemented the system on the *Tofino switch* and conducted large-scale simulations on *ns-3*
- Achieved near-zero overhead on user traffic while maintaining high sustainable throughput for telemetry data

Real-Time, Continuous Network Monitoring over Multiple Cores Sep. 2021 – Sep. 2022

- Designed a scalable software framework for sketch-based monitoring across multiple cores
- Applied the framework to nine sketch algorithms on three platforms: *CPU*, *DPDK*, and *XDPA*
- Improved accuracy by $15.6\times$ and throughput by $4.5\times$ over state-of-the-art solutions

High-Performance Measurement over Arbitrary Partial Key Query Jul. 2020 – Jul. 2021

- Developed a measurement system supporting simultaneous queries on multiple keys
- Implemented the system on *CPU*, *Open vSwitch*, *P4*, and *FPGA* platforms
- Enhanced accuracy by $10.4\times$ and throughput by $27.2\times$ in multi-key flow measurement

Approximate Streaming Algorithm

Near-optimal Global Summary for Aggregation in Wide Area Networks Jun. 2021 – Mar. 2022

- Designed a fast, adaptive, and accurate sampling scheme for global aggregation in WAN
- Derived error bounds and proved the optimality of the algorithm
- Applied the algorithm to *federated learning*, *distributed state aggregation*, and *hierarchical aggregation*

Fast and Accurate Sketch for Persistence Estimation Jan. 2020 – Sep. 2020

- Designed a sketch algorithm for persistence estimation and identifying persistent items
- Derived error bounds and demonstrated reduced space complexity
- Achieved $6.17\times$ better accuracy across multiple datasets in experimental evaluations

INDUSTRY EXPERIENCE

ByteDance | Applied Research Intern

Bellevue, Washington

Hardware Acceleration Architectures for Layer-4 Load Balancing

May 2024 – Present

- Compared two hardware acceleration architectures for stateful layer-4 load balancing.
- Conducted in-depth analysis of the design of *AMD Pensando DPU accelerators*

High-Performance Flow Logging via Matrix Sketching

May 2024 – Present

- Designed a flow logging system generates approximate summaries and supports diverse query granularities
- Leveraged *matrix sketching* theory and implemented the system on the *Tofino switch*

Conviva | Research Intern

Beijing, China

Online Profiling and Adaptation of Quality Sensitivity for Internet Video

Jul. 2020 – Jan. 2021

- Designed and implemented a data processing engine for time-state analytics
- Measured viewership patterns across 7.6M video sessions from four content providers
- Conducted in-depth analysis of the relationship between video content and quality sensitivity

MISC

Languages: C++/C, Python, CUDA, Java, SQL

Fellowship: Jonathan M. Smith Fellowship (2022)

Teaching Assistant: CIS 553: Networked Systems (Spring 2023, Fall 2023)

Shadow TPC Member: IMC 2022, EuroSys 2023, CoNEXT 2025